

Chapter 3: Population mean & covariance and Multivariate normal distribution

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Random samples from a distribution

- In order to study the sampling variability of statistics like $\bar{\mathbf{x}}$ and $\mathbf{S}_n$ with the ultimate aim of making inferences, we need to make assumptions about the variables whose observed values constitute the data set $\mathbf{X}$.
- Suppose, then, that the data have not yet been observed, but we intend to collect n sets of measurements on p variables. Consequently, we treat them as random variables.
- A random sample can now be defined. If the row vectors $\mathbf{x}_1^\top, \mathbf{x}_2^\top, \dots, \mathbf{x}_n^\top$ in $\mathbf{X} \in \mathbb{R}^{n \times p}$ represent independent observations drawn from a common joint distribution with density function $f(\mathbf{x}) = f(x_1, x_2, \dots, x_p)$, then $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ are said to form a *random sample* from $f(\mathbf{x})$.

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix}$$

$$x_i \sim f(\vec{x})$$

Expectation of random matrices(vectors)

- A random matrix/vector is a matrix/vector whose elements are random variables.
- Let $\mathbf{X} = \{X_{ij}\}$ be an $n \times p$ random matrix. Then the expected value of $\mathbf{X}$, denoted by $E(\mathbf{X})$, is the $n \times p$ matrix of numbers (if they exist)

$$E(\mathbf{X}) = \begin{bmatrix} E(X_{11}) & E(X_{12}) & \cdots & E(X_{1p}) \\ E(X_{21}) & E(X_{22}) & \cdots & E(X_{2p}) \\ \vdots & \vdots & \ddots & \vdots \\ E(X_{n1}) & E(X_{n2}) & \cdots & E(X_{np}) \end{bmatrix}.$$

- Let $\mathbf{X}$ and $\mathbf{Y}$ be random matrices of the same dimension, and let $\mathbf{A}$ and $\mathbf{B}$ be conformable matrices of constants. Then

$$E(\mathbf{X} + \mathbf{Y}) = E(\mathbf{X}) + E(\mathbf{Y}) \quad \begin{matrix} E[\text{row } A \times X \times \text{column } B] \\ = E[a_i^T X b_j] \end{matrix}$$

A is fixed matrix

$$E(\mathbf{AXB}) = \mathbf{A}E(\mathbf{X})\mathbf{B}$$

$$E(\mathbf{AXX}^T\mathbf{A}^T) = \mathbf{A}E(\mathbf{XX}^T)\mathbf{A}^T \neq \mathbf{A}E(\mathbf{X})E(\mathbf{X}^T)\mathbf{A}^T \quad \text{because } \mathbf{X} \text{ and } \mathbf{X}^T \text{ is correlated}$$

$$E(Z^2) = (E(Z))^2 \Leftrightarrow \text{Var}(Z) = 0$$

Population mean and variance

The marginal mean and variance of a random vector $\mathbf{X}^\top = [X_1, X_2, \dots, X_p]$:

$$\mu_i = \begin{cases} \int_{-\infty}^{\infty} x_i f_i(x_i) dx_i & \text{if } X_i \text{ is a continuous random variable with probability} \\ & \text{density function } f_i(x_i) \\ \sum_{\text{all } x_i} x_i p_i(x_i) & \text{if } X_i \text{ is a discrete random variable with probability} \\ & \text{function } p_i(x_i) \end{cases}$$

$$\sigma_i^2 = \begin{cases} \int_{-\infty}^{\infty} (x_i - \mu_i)^2 f_i(x_i) dx_i & \text{if } X_i \text{ is a continuous random variable} \\ & \text{with probability density function } f_i(x_i) \\ \sum_{\text{all } x_i} (x_i - \mu_i)^2 p_i(x_i) & \text{if } X_i \text{ is a discrete random variable} \\ & \text{with probability function } p_i(x_i) \end{cases}$$

It will be convenient later to denote the marginal variances by σ_{ii} .

Population covariance

The behavior of any pair of random variables, such as X_i and X_k is described by their joint probability function, and a measure of the linear association between them is provided by the covariance:

$$\sigma_{ik} = E(X_i - \mu_i)(X_k - \mu_k)$$

$$= \begin{cases} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x_i - \mu_i)(x_k - \mu_k) f_{ik}(x_i, x_k) dx_i dx_k & \text{if } X_i, X_k \text{ are continuous} \\ & \text{random variables with} \\ & \text{the joint density} \\ & \text{function } f_{ik}(x_i, x_k) \\ \sum_{\text{all } x_i} \sum_{\text{all } x_k} (x_i - \mu_i)(x_k - \mu_k) p_{ik}(x_i, x_k) & \text{if } X_i, X_k \text{ are discrete} \\ & \text{random variable with} \\ & \text{joint probability} \\ & \text{function } p_{ik}(x_i, x_k) \end{cases}$$

Independence

- More generally, the collective behavior of the p random variables $X_1, X_2, \dots, X_p$ is described by a joint probability density function $f(x_1, x_2, \dots, x_p) = f(\mathbf{x})$. ($f(\mathbf{x})$ will often be the multivariate normal density function in this course)

- If

$$P[X_i \leq x_i \text{ and } X_k \leq x_k] = P[X_i \leq x_i] P[X_k \leq x_k]$$

for all pairs of values x_i, x_k , then X_i and X_k are said to be *independent*. When X_i and X_k are continuous random variables with joint density $f_{ik}(x_i, x_k)$ and marginal densities $f_i(x_i)$ and $f_k(x_k)$, the independence condition becomes

$$f_{ik}(x_i, x_k) = f_i(x_i) f_k(x_k)$$

for all pairs (x_i, x_k) .

Independence

$X_i \in \mathbb{R}^{4 \times 1}$
 $i=1 \text{ and } 2.$

two variables
 $X_1 \quad X_2$
 $\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \\ 7 & 8 \end{pmatrix}$

- The p continuous random variables $X_1, X_2, \dots, X_p$ are *mutually independent* if their joint density can be factored as

$$f_{12\dots p}(x_1, x_2, \dots, x_p) = f_1(x_1) f_2(x_2) \cdots f_p(x_p)$$

for all p -tuples $(x_1, x_2, \dots, x_p)$.

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

- Independence of X_i and X_k implies that $\text{Cov}(X_i, X_k) = 0$.

However, the converse of is not true in general.

$$X_1 = Z \sim N(0,1) \\ X_2 = Z^2 \\ \text{Cov}(X_1, X_2) = 0 \Rightarrow X_1 \perp\!\!\!\perp X_2$$

only linear independence

more comprehensive independence

$$\text{if } X \perp\!\!\!\perp Y \Rightarrow E(g(X)) = E(g(X)|Y=y) = E(g(X))$$

$$E(g(X)h(Y)) = E(g(X))E(h(Y))$$

$$E(g(X)h(Y)) = E(g(X)|Y=y)h(y) = E(g(X))h(y) = E(g(X))E(h(Y))$$

Population mean vec and population covariance mat

We shall refer to $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ below as the *population mean* (vector) and *population covariance* (matrix), respectively:

$$\boldsymbol{\mu} = E(\mathbf{X}) = \begin{bmatrix} E(X_1) \\ E(X_2) \\ \vdots \\ E(X_p) \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{bmatrix}$$

$$\begin{aligned} \boldsymbol{\Sigma} &= E(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top \\ &= E \left(\begin{bmatrix} X_1 - \mu_1 \\ X_2 - \mu_2 \\ \vdots \\ X_p - \mu_p \end{bmatrix} [X_1 - \mu_1, X_2 - \mu_2, \dots, X_p - \mu_p] \right) \\ &= (\sigma_{ij})_{1 \leq i, j \leq p}. \end{aligned}$$

Here and below in this Chapter $\mathbf{X}$ refers to a random vector.

Unbiasedness

Handwritten derivations showing the unbiasedness of sample mean and covariance estimators.

Left side (Sample Mean):

$$\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^T$$

$$= \frac{1}{n-1} \sum_{i=1}^n (x_i - \mu + \mu - \bar{x})(x_i - \mu + \mu - \bar{x})^T$$

$$= \sum$$

Right side (Sample Covariance):

$$\begin{pmatrix} \frac{x_1^T}{x_2^T} \\ \vdots \\ x_n^T \end{pmatrix} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{np} \end{pmatrix}$$

$\text{Cov}(X_i) \in \mathbb{R}^{p \times p}$

$\text{Cov}(X_i) = \text{Cov} \begin{pmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{ip} \end{pmatrix}$

$\text{Cov}(X_i) = \text{Cov}(x_{i1}, \dots, x_{ip}) = \text{Cov}(x_{i1}, \dots, x_{ip})$

If $\mathbf{x}_1, \dots, \mathbf{x}_n$ are random samples drawn from a distribution with mean $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$, sample mean $\bar{\mathbf{x}}$ and sample covariance

$$\frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}}) (\mathbf{x}_i - \bar{\mathbf{x}})^T$$

are unbiased estimators of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$, respectively.

Correlation matrix

The *population correlation matrix*

$$\boldsymbol{\rho} = \begin{bmatrix} \frac{\sigma_{11}}{\sqrt{\sigma_{11}}\sqrt{\sigma_{11}}} & \frac{\sigma_{12}}{\sqrt{\sigma_{11}}\sqrt{\sigma_{22}}} & \cdots & \frac{\sigma_{1p}}{\sqrt{\sigma_{11}}\sqrt{\sigma_{pp}}} \\ \frac{\sigma_{12}}{\sqrt{\sigma_{11}}\sqrt{\sigma_{22}}} & \frac{\sigma_{22}}{\sqrt{\sigma_{22}}\sqrt{\sigma_{22}}} & \cdots & \frac{\sigma_{2p}}{\sqrt{\sigma_{22}}\sqrt{\sigma_{pp}}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\sigma_{1p}}{\sqrt{\sigma_{11}}\sqrt{\sigma_{pp}}} & \frac{\sigma_{2p}}{\sqrt{\sigma_{22}}\sqrt{\sigma_{pp}}} & \cdots & \frac{\sigma_{pp}}{\sqrt{\sigma_{pp}}\sqrt{\sigma_{pp}}} \end{bmatrix} = \mathbf{D}_{\sigma}^{-1/2} \boldsymbol{\Sigma} \mathbf{D}_{\sigma}^{-1/2},$$

where $\mathbf{D}_{\sigma} = \text{diag}(\sigma_{11}, \dots, \sigma_{pp})$.

Correspondingly, the sample correlation matrix can be represented by $\mathbf{D}_s^{-1/2} \mathbf{S} \mathbf{D}_s^{-1/2}$ where $\mathbf{D}_s$ is the diagonal matrix of $\mathbf{S}$.

Linear combination of random vector

random vector
 $\vec{X} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{pmatrix} \in \mathbb{R}^p$
 $E\vec{X} = \mu, \text{Cov}(\vec{X}) = \Sigma$
 Assume $x_1, \dots, x_n$ are independent
 and have same distributions as
 that of X
 for $\bar{X} = \frac{1}{n} \sum_{i=1}^n x_i, S = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^T$
 we have $E\bar{X} = \mu, ES = \Sigma$

- The linear combination $\mathbf{c}^\top \mathbf{X} = c_1 X_1 + \dots + c_p X_p$ has

$$\text{mean} = E(\mathbf{c}^\top \mathbf{X}) = \mathbf{c}^\top \boldsymbol{\mu}$$

$$\text{variance} = \text{Var}(\mathbf{c}^\top \mathbf{X}) = \mathbf{c}^\top \boldsymbol{\Sigma} \mathbf{c}$$

where $\boldsymbol{\mu} = E(\mathbf{X})$ and $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{X})$.

- Given $\mathbf{x}_1, \dots, \mathbf{x}_n$ and the sample mean $\bar{\mathbf{x}}$ and sample covariance matrices $\mathbf{S}$. What are the sample mean and sample variance of $\mathbf{c}^\top \mathbf{x}_1, \dots, \mathbf{c}^\top \mathbf{x}_n$?

• $X \in \mathbb{R}^p$ $c \in \mathbb{R}^p$ fixed find $E c^T X$ $\text{Var}(c^T X)$ Assume $E(X) = \mu$ and $\text{Cov}(X) = \Sigma$

↓
r.v.

$$E c^T X = c^T E(X) = c^T \mu$$

↓
 c is not random

$$\text{Var}(c^T X) = E(c^T X - E c^T X)(c^T X - E c^T X) = E(c^T X - E c^T X)(X^T c - E X^T c) = c^T \{E(X - EX)(X - EX)^T\} c = c^T \Sigma c = \|c\|_2^2 \Sigma$$

by above, $E(c^T X_1) = c^T E(X_1) = c^T \bar{x}$

$$\text{Var}(c^T X_1) = c^T \text{Var}(X_1) c = c^T S c = \|c\|_2^2 S$$

for $i=1$, similar for $i > 1$

$$c^T X = \sum_{i=1}^n c_i x_i$$

↓ several combination

$$CX = \begin{pmatrix} c_1^T X \\ \vdots \\ c_n^T X \end{pmatrix}$$

$$E(CX) = C E X = C \mu$$

$$\text{Cov}(CX) = C \Sigma C^T$$

result

Linear combination of random vector

In general, consider the q linear combinations of the p random variables $X_1, \dots, X_p$:

$$\mathbf{W} = \begin{bmatrix} W_1 \\ W_2 \\ \vdots \\ W_q \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1p} \\ c_{21} & c_{22} & \cdots & c_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ c_{q1} & c_{q2} & \cdots & c_{qp} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{bmatrix} = \mathbf{C}\mathbf{X}.$$

We have

$$\begin{aligned} \boldsymbol{\mu}_{\mathbf{W}} &= E(\mathbf{W}) = E(\mathbf{C}\mathbf{X}) = \mathbf{C}\boldsymbol{\mu}_{\mathbf{X}}, \\ \boldsymbol{\Sigma}_{\mathbf{W}} &= \text{Cov}(\mathbf{W}) = \text{Cov}(\mathbf{C}\mathbf{X}) = \mathbf{C}\boldsymbol{\Sigma}_{\mathbf{X}}\mathbf{C}^{\top}, \end{aligned} \tag{1}$$

where $\boldsymbol{\mu}_{\mathbf{X}}$ and $\boldsymbol{\Sigma}_{\mathbf{X}}$ are the mean vector and variance-covariance matrix of $\mathbf{X}$.

Example

Let $\mathbf{X} = [X_1, X_2]^\top$ be a random vector with mean vector $\boldsymbol{\mu}_{\mathbf{X}} = [\mu_1, \mu_2]^\top$ and variance-covariance matrix

$$\boldsymbol{\Sigma}_{\mathbf{X}} = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix}$$

Find the covariance matrix for $\mathbf{Z} = [Z_1, Z_2]^\top$ where

$$Z_1 = X_1 - X_2, \quad Z_2 = X_1 + X_2.$$

Solution: We can write

$$\mathbf{Z} = \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} := \mathbf{C}\mathbf{X}.$$

Therefore, the covariance matrix is given by

$$\begin{aligned} \text{Cov}(\mathbf{Z}) &= \mathbf{C}\mathbf{\Sigma}_\mathbf{X}\mathbf{C}^\top = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \sigma_{11} - 2\sigma_{12} + \sigma_{22} & \sigma_{11} - \sigma_{22} \\ \sigma_{11} - \sigma_{22} & \sigma_{11} + 2\sigma_{12} + \sigma_{22} \end{bmatrix} \end{aligned}$$

Cross-covariance

where $E(E(Y)^T) = E(Y)^T$

$$E\left[\left(\mathbf{X}\mathbf{Y}^T - \mathbf{X}E(\mathbf{Y})^T - E(\mathbf{X})\mathbf{Y}^T + E(\mathbf{X})E(\mathbf{Y})^T\right)\right]$$

$$= E(\mathbf{X}\mathbf{Y}^T) - E(\mathbf{X})E(\mathbf{Y})^T$$

- Cross-covariance of random vectors $\mathbf{X}$ and $\mathbf{Y}$ equals

$$\begin{aligned}\text{Cov}(\mathbf{X}, \mathbf{Y}) &= E[(\mathbf{X} - E[\mathbf{X}])(\mathbf{Y} - E[\mathbf{Y}])^T] \\ &= E[\mathbf{X}\mathbf{Y}^T] - E[\mathbf{X}]E[\mathbf{Y}]^T.\end{aligned}$$

- Let $\mathbf{A}$, $\mathbf{B}$ be conformable matrices of constants, then

$$\text{Cov}(\mathbf{A}\mathbf{X}, \mathbf{B}\mathbf{Y}) = \mathbf{A} \text{Cov}(\mathbf{X}, \mathbf{Y}) \mathbf{B}^T.$$

- $\text{Cov}(\mathbf{C}\mathbf{X}) = \mathbf{C}\Sigma_X\mathbf{C}^T$ is a special case of the above.

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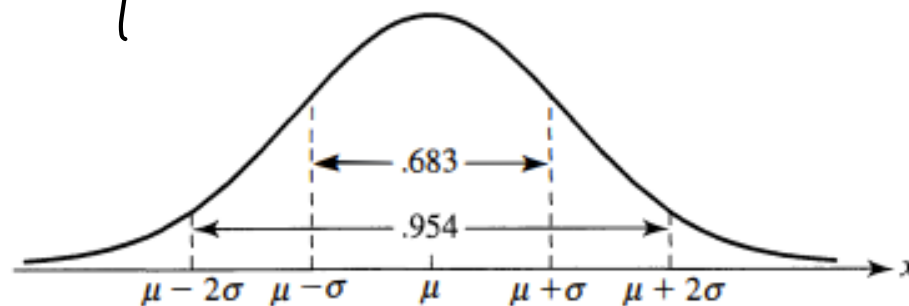
- 1 Population mean and covariance matrices
- 2 Multivariate Normal distribution

(Univariate) normal distribution

Univariate Gaussian (normal) density:

$$\begin{aligned} f_X(x) &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)} \\ &= \frac{1}{(2\pi)^{\frac{1}{2}} (\sigma^2)^{\frac{1}{2}}} e^{-\frac{1}{2}(x-\mu) \frac{1}{\sigma^2} (x-\mu)} \end{aligned}$$

$p(X=a) = 0$ in $f_X(x)$ is continuous or $X \sim \mathcal{N}(\mu, \sigma^2)$



but here $P(\mu - \sigma \leq X \leq \mu + \sigma) = 0.683$

Figure 1: A normal density with mean μ and variance σ^2 and selected areas under the curve
which means the cumulative P is higher around μ

Multivariate normal density

Definition 1

A random vector $\mathbf{X}$ has a Multivariate Normal (MVN) distribution if it has a joint probability density function (pdf) of the form

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{p}{2}} |\boldsymbol{\Sigma}|^{\frac{1}{2}}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\top} \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}$$

$\downarrow$
 determinant

mean and covariance

We denote this p -dimensional normal density by $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, which is analogous to the normal density in the univariate case.

Special cases: diagonal Σ

For p-variate case, if Σ is diagonal, then

$$\Sigma^{-1} = \begin{bmatrix} \frac{1}{\sigma_{11}} & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \frac{1}{\sigma_{pp}} \end{bmatrix} \quad \text{and} \quad \underline{|\Sigma| = (\sigma_{11}) (\sigma_{22}) \cdots (\sigma_{pp})}$$

$$\begin{aligned} \text{so } f_{\mathbf{x}}(\mathbf{x}) &= \frac{1}{(2\pi)^{\frac{p}{2}} \sqrt{(\sigma_{11}) \cdots (\sigma_{pp})}} \\ &\quad \times \exp \left\{ -\frac{1}{2} \frac{(x_1 - \mu_1)^2}{\sigma_{11}} - \cdots - \frac{1}{2} \frac{(x_p - \mu_p)^2}{\sigma_{pp}} \right\} \\ &= f_{x_1}(x_1) \cdot f_{x_2}(x_2) \cdots f_{x_p}(x_p). \end{aligned}$$

they are independent to each other

Bivariate case

- For bivariate case,

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \sim N_2 \left[\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_{11}^2 & \rho_{12} \sigma_{11} \sigma_{22} \\ \rho_{12} \sigma_{11} \sigma_{22} & \sigma_{22}^2 \end{pmatrix} \right].$$

- If

$$\rho_{12} = 0, f_{\mathbf{x}} \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right) = f_{x_1}(x_1) \cdot f_{x_2}(x_2).$$

$$B \sim N(\mu, \sigma^2)$$

Shape of the density

Two bivariate distributions are shown below.

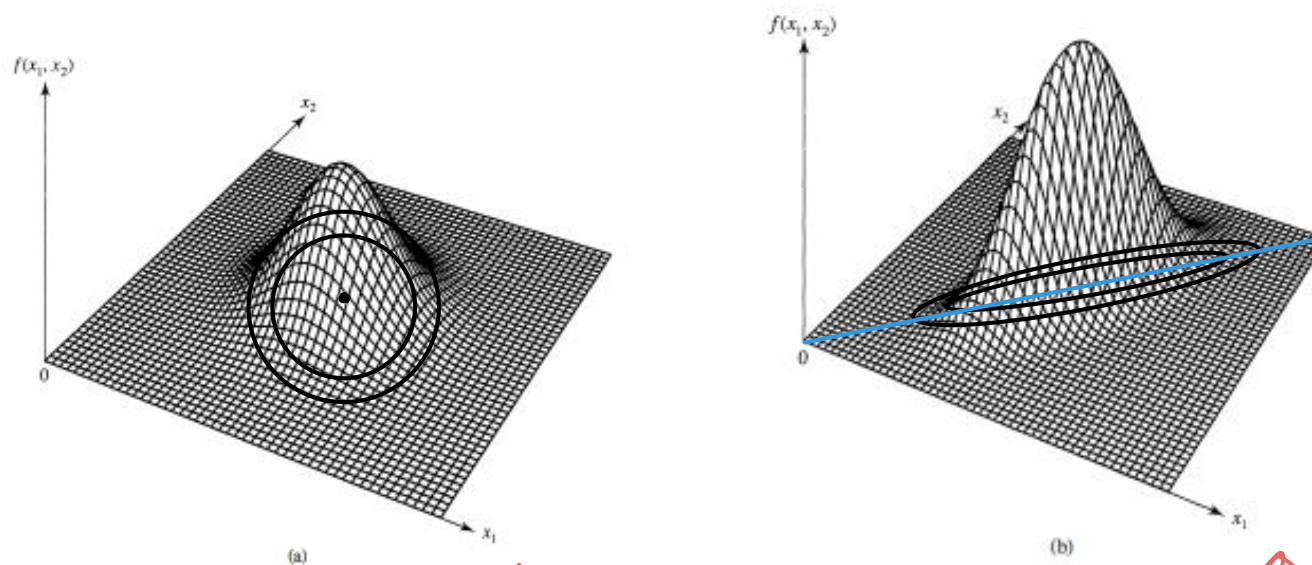


Figure 2: Two bivariate normal distributions. (a) $\sigma_{11} = \sigma_{22}$ and $\rho_{12} = 0$. (b) $\sigma_{11} = \sigma_{22}$ and $\rho_{12} = .75$.

Handwritten notes:

$$|\Sigma_1|^{\frac{1}{2}} \neq |\Sigma_2|^{\frac{1}{2}}$$

$$\Sigma_1 = \begin{bmatrix} \sigma_{11} & 0 \\ 0 & \sigma_{22} \end{bmatrix} \quad \Sigma_2 = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}$$

Annotations on the figure:

- Red arrows point from the text $|\Sigma_1|^{\frac{1}{2}} \neq |\Sigma_2|^{\frac{1}{2}}$ to the two plots, indicating that the peak heights are different.
- Blue text: $\therefore f_2(x)_{\max} > f_1(x)_{\max}$
- Blue text: $\therefore |\Sigma_2| = \sigma_{11}\sigma_{22} - \sigma_{12}^2 < |\Sigma_1|$

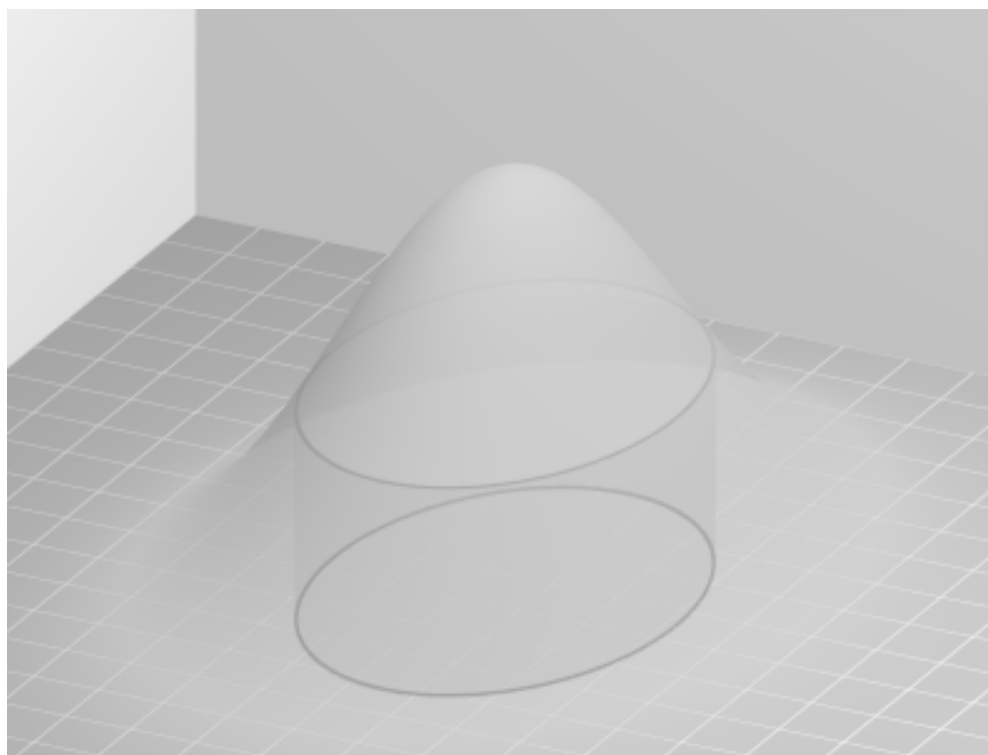
Constant density contour

$$f(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} e^{-\frac{(\mathbf{x}-\mu)^\top \Sigma^{-1} (\mathbf{x}-\mu)}{2}} = d$$

$(\mathbf{x}-\mu)^\top \Sigma^{-1} (\mathbf{x}-\mu) = 2 \log \left(\frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \right) = \text{Constant}$

Values of $\mathbf{x}$ yielding constant height for density are ellipsoids.

$$\text{Constant density contour} = \{ \mathbf{x} : (\mathbf{x} - \mu)^\top \Sigma^{-1} (\mathbf{x} - \mu) = c^2 \}$$



$$P(x_1 \in [a, b]) = \text{Area under curve}$$

in univariate case
area represent P

in multivariate case
volume represent P

$$P((x_1, x_2) \in O) = \text{Volume under surface}$$

Figure 3: Constant density contour for bivariate normal

Constant density contour

$$\sigma_{11} > \sigma_{22} \text{ and } \sigma_{12} = 0$$

$$\Sigma = \begin{pmatrix} \sigma_{11} & 0 \\ 0 & \sigma_{22} \end{pmatrix}$$

The constant density contours are centered at μ .

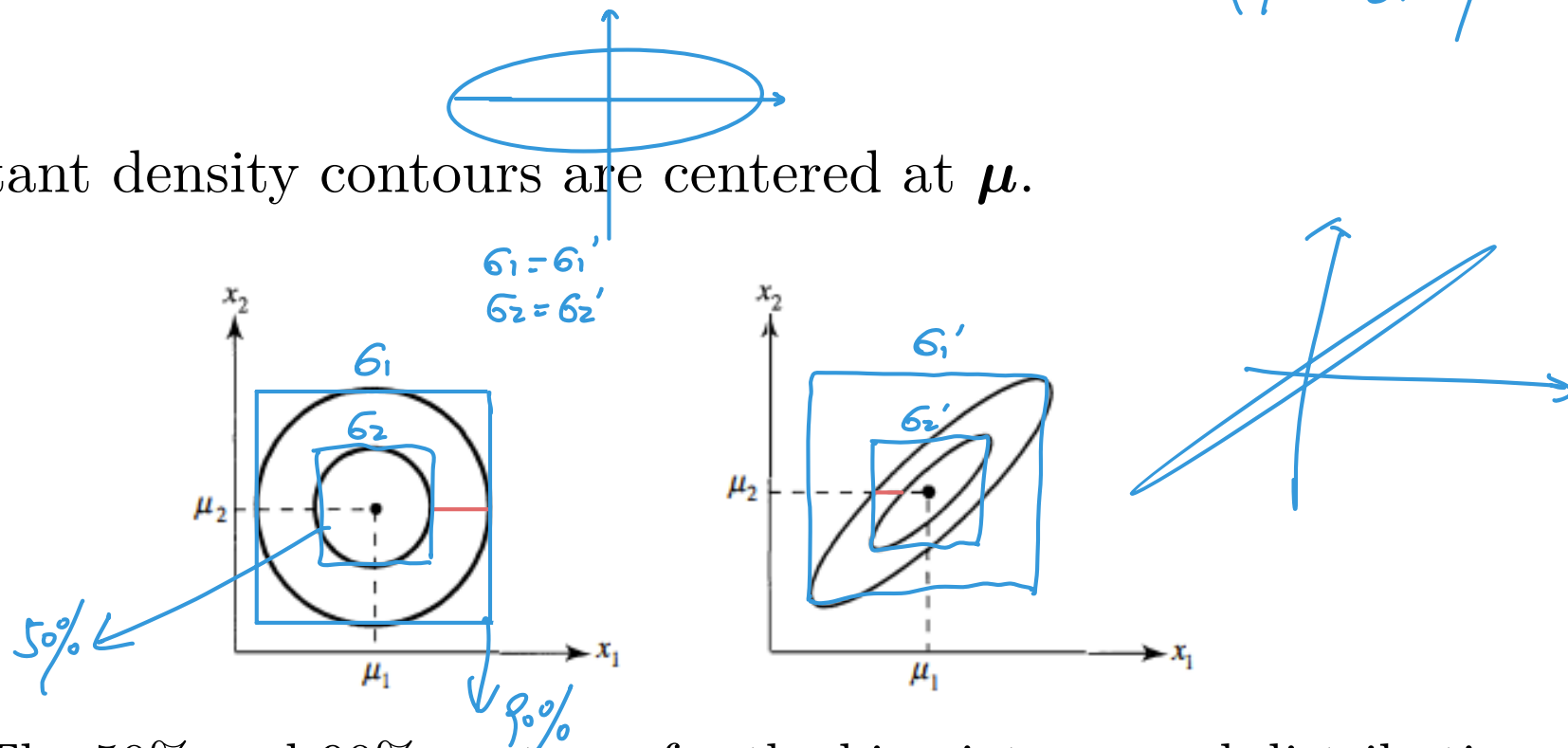


Figure 4: The 50% and 90% contours for the bivariate normal distributions in Figure 2

Some equivalent definitions of MVN

Besides specifying the density to define the multivariate normal distribution, we have some equivalent definitions:

Definition 2

$$Z_i \sim \mathcal{N}(0, 1)$$

linear combination of Z will be multivariate normal

A random vector $\mathbf{X}$ has a multivariate normal distribution if there exist $\boldsymbol{\mu} \in \mathbb{R}^k$, $\mathbf{A} \in \mathbb{R}^{k \times \ell}$ such that $\mathbf{X} = \mathbf{A}\mathbf{Z} + \boldsymbol{\mu}$ where $\mathbf{Z}$ is a random vector whose components are independent $\mathcal{N}(0, 1)$ random variables.

Definition 3

If for every $\mathbf{a}$, $\mathbf{a}^T \mathbf{X}$ is normal then $\mathbf{C}\mathbf{X}$ is normal because $\mathbf{a}^T \mathbf{C}\mathbf{X} = (\mathbf{C}^T \mathbf{a})^T \mathbf{X}$ is normal by df. 3 and $\mathbf{X} \sim \text{MVN}$

A random vector $\mathbf{X}$ has a multivariate normal distribution if for every real vector $\mathbf{a}$, the random variable $\mathbf{a}^T \mathbf{X}$ is normal.

If $\mathbf{X} \sim \text{MVN}$, then $\mathbf{C}\mathbf{X}$ is MVN
 we can find $\mathbf{A}, \boldsymbol{\mu}$, s.t. $\mathbf{C}\mathbf{X} = \mathbf{A}\mathbf{Z} + \boldsymbol{\mu}$ $\mathbf{Z} = \begin{pmatrix} Z_1 \\ \vdots \\ Z_p \end{pmatrix}$ $Z_i \text{ iid } \mathcal{N}(0, 1)$

$$CX = CAZ + C\mu \quad \text{by df. 2 it is MVN}$$

Some important properties for multivariate normal vectors

- ① Linear combinations of the components of $\mathbf{X}$ are normally distributed.
- ② All subsets of the components of $\mathbf{X}$ have a (multivariate) normal distribution. *linear combination of original vectors*
- ③ Zero covariance implies that the corresponding components are independently distributed.
- ④ The conditional distributions of the components are (multivariate) normal.

Property 1: Linear combinations of a normal vector is normal

If $X \sim N_p(\mu, \Sigma)$
 then $CX + d \sim N_p(C\mu + d, C\Sigma C^T)$

If $\mathbf{X}$ is distributed as $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the q linear combinations

if $C \in \mathbb{R}^{q \times p}$

$$\underset{(q \times p)}{\mathbf{A}} \underset{(p \times 1)}{\mathbf{X}} = \begin{bmatrix} a_{11}X_1 + \cdots + a_{1p}X_p \\ a_{21}X_1 + \cdots + a_{2p}X_p \\ \vdots \\ a_{q1}X_1 + \cdots + a_{qp}X_p \end{bmatrix}$$

are distributed as $N_q(\mathbf{A}\boldsymbol{\mu}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T)$. Also, $\underset{(p \times 1)}{\mathbf{X}} + \underset{(p \times 1)}{\mathbf{d}}$, where $\mathbf{d}$ is a vector of constants, is distributed as $N_p(\boldsymbol{\mu} + \mathbf{d}, \boldsymbol{\Sigma})$.

Example

For $\mathbf{X} = (X_1, X_2, X_3)^\top$ distributed as $N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, find the distribution of $(X_1 - X_2, X_2 - X_3)^\top$.

Example

For $\mathbf{X} = (X_1, X_2, X_3)^\top$ distributed as $N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, find the distribution of $(X_1 - X_2, X_2 - X_3)^\top$.

We write

$$\begin{bmatrix} X_1 - X_2 \\ X_2 - X_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \mathbf{A}\mathbf{X}$$

Example

Then the distribution of $\mathbf{AX}$ is multivariate normal with mean

$$\mathbf{A}\boldsymbol{\mu} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix} = \begin{bmatrix} \mu_1 - \mu_2 \\ \mu_2 - \mu_3 \end{bmatrix}$$

and covariance matrix

$$\begin{aligned} \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top &= \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_{22} & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{bmatrix} \\ &= \begin{bmatrix} \sigma_{11} - 2\sigma_{12} + \sigma_{22} & \sigma_{12} + \sigma_{23} - \sigma_{22} - \sigma_{13} \\ \sigma_{12} + \sigma_{23} - \sigma_{22} - \sigma_{13} & \sigma_{22} - 2\sigma_{23} + \sigma_{33} \end{bmatrix} \end{aligned}$$

Property 2: Subsets of components of normal are normal

$\mathbf{X} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{pmatrix}$ is MVN, then $\begin{pmatrix} x_3 \\ x_5 \end{pmatrix}$ is MVN.
 because $\begin{pmatrix} x_3 \\ x_5 \end{pmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 0 & 1 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix}$

All subsets of $\mathbf{X}$ are normally distributed. If we respectively partition $\mathbf{X}$, its mean vector $\boldsymbol{\mu}$, and its covariance matrix $\boldsymbol{\Sigma}$ as

$\text{Cov} \begin{pmatrix} x_3 \\ x_5 \end{pmatrix} = \begin{bmatrix} 6_{33} & 6_{35} \\ 6_{55} & 6_{55} \end{bmatrix}$

$$\underset{(p \times 1)}{\mathbf{X}} = \begin{bmatrix} \underset{(q \times 1)}{\mathbf{X}_1} \\ \underset{((p-q) \times 1)}{\mathbf{X}_2} \end{bmatrix} \quad \underset{(p \times 1)}{\boldsymbol{\mu}} = \begin{bmatrix} \underset{(q \times 1)}{\boldsymbol{\mu}_1} \\ \underset{((p-q) \times 1)}{\boldsymbol{\mu}_2} \end{bmatrix}$$

and

$$\underset{(p \times p)}{\boldsymbol{\Sigma}} = \begin{bmatrix} \underset{(q \times q)}{\boldsymbol{\Sigma}_{11}} & \underset{(q \times (p-q))}{\boldsymbol{\Sigma}_{12}} \\ \underset{((p-q) \times q)}{\boldsymbol{\Sigma}_{21}} & \underset{((p-q) \times (p-q))}{\boldsymbol{\Sigma}_{22}} \end{bmatrix}$$

then $\mathbf{X}_1$ is distributed as $N_q(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$.

Property 3

If $\begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$ is MVN, $\text{Cov}(X_1, X_2) = 0$, then $X_1 \perp X_2$

because $\begin{pmatrix} \Sigma_{11} & 0 \\ 0 & \Sigma_{22} \end{pmatrix}$ is $\text{Cov}(X_1)$ and $\text{Cov}(X_2)$

$$\textcircled{2} (X - \mu)^T \Sigma^{-1} (X - \mu)$$

$$= (X_1 - \mu_1, X_2 - \mu_2) \begin{pmatrix} \Sigma_1^{-1} & 0 \\ 0 & \Sigma_2^{-1} \end{pmatrix} \begin{pmatrix} X_1 - \mu_1 \\ X_2 - \mu_2 \end{pmatrix}$$

$$\text{joint density} = \frac{1}{(2\pi)^{\frac{p}{2}} |\Sigma|^{\frac{1}{2}}} e^{-\frac{(X-\mu)^T \Sigma^{-1} (X-\mu)}{2}} = \frac{1}{(2\pi)^{\frac{k}{2}} |\Sigma_1|^{\frac{1}{2}}} e^{-\frac{(X_1-\mu_1)^T \Sigma_1^{-1} (X_1-\mu_1)}{2}} \times \frac{1}{(2\pi)^{\frac{p-k}{2}} |\Sigma_2|^{\frac{1}{2}}} e^{-\frac{(X_2-\mu_2)^T \Sigma_2^{-1} (X_2-\mu_2)}{2}}$$

here $|\Sigma| = |\Sigma_1| \cdot |\Sigma_2|$

$\Leftarrow$ generally not true.

- If $\mathbf{X}_1 \in \mathbb{R}^{q_1}$ and $\mathbf{X}_2 \in \mathbb{R}^{q_2}$ are independent, then $\text{Cov}(\mathbf{X}_1, \mathbf{X}_2) = \mathbf{0}$, a $q_1 \times q_2$ matrix of zeros.

- If $\begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{bmatrix}$ is $N_{q_1+q_2} \left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \right)$, then $\mathbf{X}_1$ and $\mathbf{X}_2$ are independent if and only if $\Sigma_{12} = \mathbf{0}$.

- Note that if two normal vectors have a covariance of zero, it does not imply that they are independent. A counterexample is by taking $\mathbf{X}_2 = Z\mathbf{X}_1$ for some Z independent of $\mathbf{X}_1$ and has distribution $P(Z = \pm 1) = 1/2$.

If X_1 is MVN and X_2 is MVN and $\text{Cov}(X_1, X_2) = 0$ then $X_1 \perp X_2$ ~~X~~

$$\text{Cov} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

$\mathbb{R}^k$ $\mathbb{R}^{p-k}$

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} X_1 \end{bmatrix}^T + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} X_2 \end{bmatrix}^T$$

Example

Let $\underset{(3 \times 1)}{\mathbf{X}}$ be $N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\vec{X} \in \mathbb{R}^{3 \times 1} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$ $\text{Cov}(X) = \boldsymbol{\Sigma}$

$$\boldsymbol{\Sigma} = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$\text{Cov}(X_1, X_2) = \text{Cov}(X_2, X_1) = 1 \neq 0$
 Are X_1 and X_2 independent? What about (X_1, X_2) and X_3 ?

$$\text{Cov}\left(\begin{pmatrix} X_1 \\ X_2 \end{pmatrix}, X_3\right) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \text{Cov}(X_1, X_3) \\ \text{Cov}(X_2, X_3) \end{pmatrix}$$

and (X_1, X_2) and X_3 are jointly normal
 because $\vec{X}$ is $N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

Example

Let $\underset{(3 \times 1)}{\mathbf{X}}$ be $N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with

$$\boldsymbol{\Sigma} = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Are X_1 and X_2 independent? What about (X_1, X_2) and X_3 ?
 X_1 and X_2 are not independent; (X_1, X_2) and X_3 are independent.

Property 4: Conditional distributions of components are normal

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \text{ is MVN}$$

$$\text{then } X_1 | X_2 = x_2 \text{ is MVA}$$

$$X_2 | X_1 = x_1 \text{ is MVN}$$

Let $\mathbf{X} = \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{bmatrix}$ be distributed as $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$,

$\boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix}$, and $|\boldsymbol{\Sigma}_{22}| > 0$. Then the conditional distribution of $\mathbf{X}_1$, given that $\mathbf{X}_2 = \mathbf{x}_2$, is normal and has

$$\text{Mean} = \mu_1 + \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} (\mathbf{x}_2 - \mu_2)$$

and

$$X_1 | X_2 = x_2 \quad E(\dots)$$

$$EX = \begin{pmatrix} EX_1 \\ EX_2 \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \text{Cov} X = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

Let $Y = AX$ where $A = \begin{bmatrix} I_p & -\Sigma_{12}\Sigma_{22}^{-1} \\ 0 & I_{p-q} \end{bmatrix}$, $Y = \begin{pmatrix} X_1 - \Sigma_{12}\Sigma_{22}^{-1}X_2 \\ X_2 \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_1 - \Sigma_{12}\Sigma_{22}^{-1}\mu_2 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} & 0 \\ 0 & \Sigma_{22} \end{pmatrix}\right)$

$X_1 - \Sigma_{12}\Sigma_{22}^{-1}X_2 \checkmark$
 $X_1 - \Sigma_{12}\Sigma_{22}^{-1}X_2 | X_2 = x_2$
 then we know
 $X_1 - \Sigma_{12}\Sigma_{22}^{-1}X_2 | X_2 = x_2$

Covariance = $\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$

$E[X_1 | X_2 = x_2] = \mu_1 - \Sigma_{12}\Sigma_{22}^{-1}\mu_2 + \Sigma_{12}\Sigma_{22}^{-1}x_2$

So $X_1 - \Sigma_{12}\Sigma_{22}^{-1}X_2 \perp X_2$

Distribution of Sample mean

For any $\mathbf{a} \in \mathbb{R}^p$, $\mathbf{a}^T \bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{a}^T \mathbf{x}_i$ is normal

So $\bar{\mathbf{x}}$ is normal

$$E \bar{\mathbf{x}} = \boldsymbol{\mu}$$

- If $\mathbf{x}_i \stackrel{i.i.d.}{\sim} \mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then for any positive integer n ,

$$\text{Cov}(\bar{\mathbf{x}}) = \text{Cov}\left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i\right) = \frac{1}{n} \boldsymbol{\Sigma} \quad \uparrow \downarrow \quad \text{remain same}$$

$$\sqrt{n}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \sim N_p(0, \boldsymbol{\Sigma}).$$

$$\bar{\mathbf{x}} - \boldsymbol{\mu} \sim N_p(0, \frac{1}{n} \boldsymbol{\Sigma}) \quad \therefore \sqrt{n}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \sim N_p(0, \boldsymbol{\Sigma})$$

(Proof hint: Use Definition 3, show that $\mathbf{c}^T \bar{\mathbf{x}}$ is normal first.)

- For real data, the multivariate normal assumption may be too strong. Fortunately, the Central Limit Theorem implies that the above result still holds asymptotically as the sample size n approaches infinity.

Central Limit Theorem

(Multivariate) Central Limit Theorem

Let $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ be independent observations from any population with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$. Then

$$\sqrt{n}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \xrightarrow{d} N_p(0, \boldsymbol{\Sigma}) \text{ as } n \rightarrow \infty,$$

where convergence in distribution $\xrightarrow{d}$ means that for any set $\Omega \in \mathbb{R}^p$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(\mathbf{S}_n \in \Omega) = \mathbb{P}(\mathbf{Y} \in \Omega)$$

for a random vector $\mathbf{Y}$ with normal distribution $N_p(0, \boldsymbol{\Sigma})$.

Why MVN

Why emphasis the MVN?

- Genuinely good population model for some natural phenomena
- Even for nonnormal data, MVN is often useful approximation - such as for inferences involving sample mean vectors, which are asymptotically normal due to CLT *last page*
- Theoretical elegance, such as
 - MVN is completely characterized by its mean and covariance
 - Linear transformations of MVN are normal
 - Uncorrelated components of MVN are independent

Maximum Likelihood Estimation

- Maximum likelihood estimation (MLE) selects parameter values that maximize the joint density of the observed data.
- In the next few pages, we derive the MLE for $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ for multivariate normal data $\mathbf{x}_1, \dots, \mathbf{x}_n \stackrel{i.i.d.}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.
- This optimization leads to the MLEs: $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$ and $\hat{\boldsymbol{\Sigma}} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$.

Maximum likelihood estimation of μ and Σ

Let $\mathbf{x}_1, \dots, \mathbf{x}_n$ be a random sample from $N_p(\mu, \Sigma)$.

Joint density:

$$\begin{aligned}
 f(\mathbf{x}_1, \dots, \mathbf{x}_n) &= \prod_{i=1}^n f(\mathbf{x}_i) \\
 &= \prod_{i=1}^n \frac{1}{(2\pi)^{\frac{p}{2}} |\Sigma|^{\frac{1}{2}}} \exp \left\{ -\frac{1}{2} (\mathbf{x}_i - \mu)^\top \Sigma^{-1} (\mathbf{x}_i - \mu) \right\} \\
 &= \frac{1}{(2\pi)^{\frac{np}{2}} |\Sigma|^{\frac{n}{2}}} \exp \left\{ -\frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \mu)^\top \Sigma^{-1} (\mathbf{x}_i - \mu) \right\} \\
 &= L(\mu, \Sigma) = L(\mathbf{x}_1, \dots, \mathbf{x}_n)
 \end{aligned}$$

Goal: Find values of μ and Σ that maximize the likelihood of observing $\mathbf{x}_1, \dots, \mathbf{x}_n$.

$$\hat{\mu}, \hat{\Sigma} = \underset{\mu, \Sigma}{\operatorname{argmax}} L(\mathbf{x}_1, \dots, \mathbf{x}_n)$$

$$\hat{\mu} = \underset{\mu}{\operatorname{argmin}} \sum_{i=1}^n (\mathbf{x}_i - \mu)^\top \Sigma^{-1} (\mathbf{x}_i - \mu)$$

Use

$$\begin{aligned}
 (\mathbf{x}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}) &= \text{tr} \left\{ (\mathbf{x}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}) \right\} \\
 &= \text{tr} \left\{ \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}) (\mathbf{x}_i - \boldsymbol{\mu})^\top \right\},
 \end{aligned}$$

$= \text{tr} \sum_{i=1}^n (x_i - \mu)^\top \Sigma^{-1} (x_i - \mu) = \text{tr} \sum_{i=1}^n \Sigma^{-1} (x_i - \mu) (x_i - \mu)^\top$
 $= \text{tr} \left[\sum_{i=1}^n \Sigma^{-1} (x_i - \bar{x} + \bar{x} - \mu) (x_i - \bar{x} + \bar{x} - \mu)^\top \right]$
 $= \text{tr} \left[\sum_{i=1}^n \Sigma^{-1} (x_i - \bar{x}) (x_i - \bar{x})^\top + n \Sigma^{-1} (\bar{x} - \mu) (\bar{x} - \mu)^\top \right]$

rewrite

$$\begin{aligned}
 \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}) &= \text{tr} \left\{ \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}) \right\} \\
 &= \text{tr} \left\{ \boldsymbol{\Sigma}^{-1} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu}) (\mathbf{x}_i - \boldsymbol{\mu})^\top \right\} \\
 &= \text{tr} \left\{ \boldsymbol{\Sigma}^{-1} \sum_{i=1}^n [(\mathbf{x}_i - \bar{\mathbf{x}}) + (\bar{\mathbf{x}} - \boldsymbol{\mu})] [(\mathbf{x}_i - \bar{\mathbf{x}}) + (\bar{\mathbf{x}} - \boldsymbol{\mu})]^\top \right\} \\
 &= \text{tr} \left\{ \boldsymbol{\Sigma}^{-1} \left[\sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}}) (\mathbf{x}_i - \bar{\mathbf{x}})^\top + \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}}) (\bar{\mathbf{x}} - \boldsymbol{\mu})^\top + \right. \right. \\
 &\quad \left. \left. \sum_{i=1}^n (\bar{\mathbf{x}} - \boldsymbol{\mu}) (\mathbf{x}_i - \bar{\mathbf{x}})^\top + n(\bar{\mathbf{x}} - \boldsymbol{\mu}) (\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \right] \right\}
 \end{aligned}$$

So,

$$L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{\frac{np}{2}} |\boldsymbol{\Sigma}|^{\frac{n}{2}}} \times \exp \left\{ -\frac{1}{2} \operatorname{tr} \left(\boldsymbol{\Sigma}^{-1} \left[\sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}}) (\mathbf{x}_i - \bar{\mathbf{x}})^\top + n(\bar{\mathbf{x}} - \boldsymbol{\mu})(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \right] \right) \right\}$$

Note that the value of $\boldsymbol{\mu}$ maximizing $L(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is the value minimizing $\operatorname{tr} \{ n\boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu})(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \}$. Since $\boldsymbol{\Sigma}^{-1}$ is positive definite,

$$\operatorname{tr} \{ n\boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu})(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \} = n(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) > 0$$

unless

$$\boldsymbol{\mu} = \bar{\mathbf{x}}.$$

Thus, the likelihood is maximized with respect to $\boldsymbol{\mu}$ at $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$.

It remains to maximize

$$L(\hat{\boldsymbol{\mu}}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{np/2} |\boldsymbol{\Sigma}|^{n/2}} e^{-\text{tr}[\boldsymbol{\Sigma}^{-1} (\sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top)]/2}$$

over $\boldsymbol{\Sigma}$.

Result: Given a $p \times p$ symmetric positive definite ^{but important} matrix $\mathbf{B}$ and a scalar $b > 0$, it follows that

$$\frac{1}{|\boldsymbol{\Sigma}|^b} e^{-\text{tr}(\boldsymbol{\Sigma}^{-1} \mathbf{B})/2} \leq \frac{1}{|\mathbf{B}|^b} (2b)^{pb} e^{-bp}$$

for all positive definite $\boldsymbol{\Sigma}$, with equality holding only for $\boldsymbol{\Sigma} = (1/2b)\mathbf{B}$.

By the above result with $b = n/2$ and $\mathbf{B} = \sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top$, the maximum occurs at $\hat{\boldsymbol{\Sigma}} = (1/n) \sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top = (n-1)\mathbf{S}/n$.

$$E[X] = \mu, \text{ Cov}(X) = \Sigma \quad X \in \mathbb{R}^p$$

$$\text{then } E[AX + b] = ? \quad A \in \mathbb{R}^{n \times p}$$

$$\text{Cov}(AX) = ?$$

MVN of

$$\textcircled{2} Ax + b \text{ with } \begin{pmatrix} \mathcal{N}(0, 1) \\ \vdots \\ \mathcal{N}(0, 1) \end{pmatrix}$$

$$\textcircled{3} b^T X \sim \text{Normal for any } b$$

Unloaded prof is not important.

MLE for μ and Σ given n i.i.d samples,